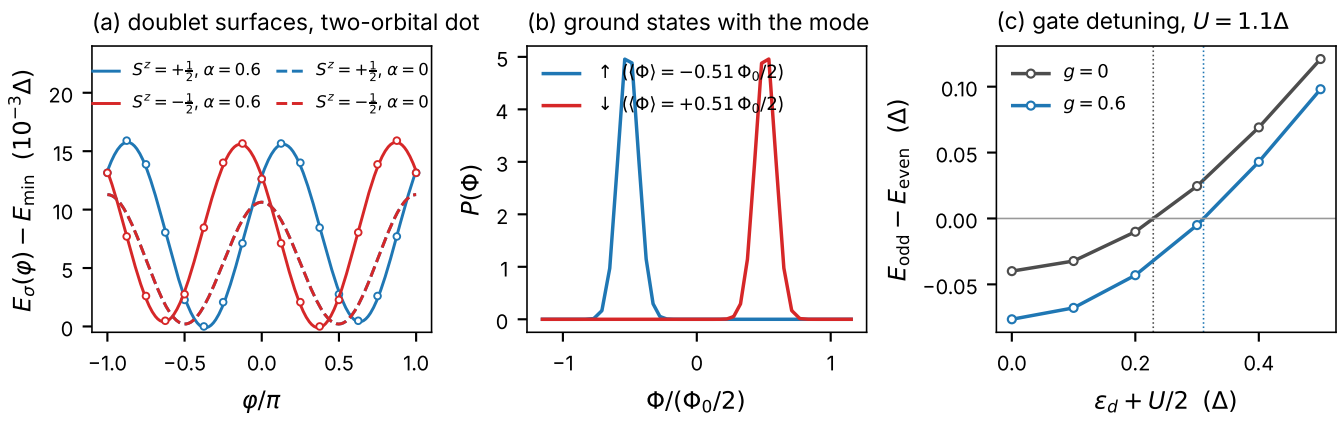
(a) doublet surfaces, two-orbital dot
$S^z = +\frac{1}{2}$, $\alpha = 0.6$
$S^z = +\frac{1}{2}$, $\alpha = 0$
$S^z = -\frac{1}{2}$, $\alpha = 0.6$
$S^z = -\frac{1}{2}$, $\alpha = 0$
$E_\sigma(\varphi) - E_{\min}$ $(10^{-3}\Delta)$
φ/π
(b) ground states with the mode
$\uparrow$ ($\langle\Phi\rangle = -0.51\,\Phi_0/2$)
$\downarrow$ ($\langle\Phi\rangle = +0.51\,\Phi_0/2$)
$P(\Phi)$
$\Phi/(\Phi_0/2)$
(c) gate detuning, $U = 1.1\Delta$
$g = 0$
$g = 0.6$
$E_{\text{odd}} - E_{\text{even}}$ (Δ)
$\varepsilon_d + U/2$ (Δ)